

Scoring, Signals, and Next Actions (Enhanced)

This document expands on the **scoring and buy decision logic**, explains folder structure, and clarifies how the `reason` field now includes **both numerical stats and human-friendly interpretations** for each indicator.

Folder Structure — Scoring Files

```
backend/app/repos/parquet/
├── datasets.py           # Defines datasets & partitions
├── indicators_repo.py    # Reads indicators from OHLC parquet
├── scores_repo.py        # Writes scores and related parquet outputs
├── scores/
│   ├── daily/            # Daily (EOD) scores – official swing-trade
│   └── dataset
│       ├── as_of=YYYY-MM-DD/
│       │   ├── part-0000.parquet
│       │   ├── part-0001.parquet
│       │   └── ...
│       └── ...
│   └── intraday/         # Intraday scores used for watchlists & live
└── heatmaps
    ├── date=YYYY-MM-DD/
    │   ├── part-0000.parquet
    │   └── ...
    └── ...
```

Notes: - The **EOD model** reads historical daily OHLC parquet files and computes indicators such as RSI, ADX, ATR10, RelVol20, and proximity to 52W high. - Intraday parquet files are refreshed by the scheduler every 15 minutes for watchlist updates. - **Base length (bars)** is derived from historical daily parquet — it counts how many daily bars (trading days) a stock has moved sideways before a breakout attempt.

Scoring Overview

Pillar	Components	Weight
Momentum	RSI, ADX, DI+/DI-, ADX slope	35%
Breakout Quality	52W proximity, higher highs/lows, EMA cluster, ATR normalization	30%
Accumulation & Volume	RelVol, OBV, CMF, volume expansion	25%

Pillar	Components	Weight
Market/Sector Context	Index trend, sector breadth	10%

Final score = weighted sum of normalized 0–100 pillar scores.

Basic Model (0–12 → %)

A lighter additive model (RSI zone + ADX band + breakout markers). Final = (points / 12) × 100.

BUY Decision Logic

BUY = Yes only if **all** EOD gates pass:

Factor	Rule	Comment
Score	≥ 70	Strong overall quality
52W Proximity	≥ -8%	Near highs
Pivot Clear %	Between +1% and +5%	Clean breakout, not overextended
Base Length	≥ 15 bars	Mature consolidation
RelVol20	≥ 1.5×	Volume confirmation
ADX14	≥ 22	Trend strength confirmed
ATR10%	Between 3–7%	Tradable volatility range
Day % Change	≤ +6%	Avoid euphoric spikes
Liquidity	≥ ₹5 Crore	Ensure safe tradability

If any condition fails → **BUY = No**, with `reason` containing both **numeric stats** and **layman summary** for each metric.

`reason` Field — Format and Meaning

Combined Format Example

```
reason: "pivot_clear:+0.3% (testing breakout) | base_len:2 (no clear base) |
RelVol:0.6x (quiet, low volume) | ATR10:2.1% (low volatility) | ADX:18 (weak
trend) | prox52:-12% (far from highs)"
```

Each metric is displayed as:

value (common-language meaning) — making it intuitive even for non-technical readers.

Components

Metric	Value Format	Meaning
<code>pivot_clear</code>	% above or below breakout	Shows if stock cleared resistance (1–5% = good)
<code>base_len</code>	number of daily bars	Duration of sideways base (≥ 15 = mature)
<code>RelVol</code>	relative volume vs 20-day average	1.5x+ = strong buying interest
<code>ATR10</code>	10-day ATR as % of price	3–7% = optimal volatility for swing trade
<code>ADX</code>	Average Directional Index	≥ 22 = solid trend, < 20 = weak trend
<code>prox52</code>	% distance from 52W high	–8% or better = near breakout zone

Human Mapping — Range → Common Language

Metric	Range	Meaning (for reason text)
pivot_clear %	$< 0 \rightarrow$ (below resistance), $0-1 \rightarrow$ (testing breakout), $1-5 \rightarrow$ (clean breakout), $> 5 \rightarrow$ (overextended)	
base_len (bars)	$< 10 \rightarrow$ (no clear base), $10-14 \rightarrow$ (early consolidation), $15-25 \rightarrow$ (ready to break), $> 30 \rightarrow$ (stale setup)	
RelVol	$< 1.0x \rightarrow$ (quiet, no strong buying), $1.0-1.4x \rightarrow$ (average volume), $1.5-2.0x \rightarrow$ (strong buying), $> 2.0x \rightarrow$ (high-volume breakout)	
ATR10 %	$< 3\% \rightarrow$ (low volatility), $3-7\% \rightarrow$ (healthy tradable), $> 7\% \rightarrow$ (too volatile)	
ADX	$< 20 \rightarrow$ (weak/sideways), $20-25 \rightarrow$ (emerging trend), $25-35 \rightarrow$ (strong trend), $> 35 \rightarrow$ (euphoric)	
prox52 %	$< -15 \rightarrow$ (far from highs), $-15--8 \rightarrow$ (building base), $-8-0 \rightarrow$ (near breakout zone), $> 0 \rightarrow$ (at highs)	

Output Behavior

• BUY = Yes:

```
reason: "pivot_clear:+2.4% (clean breakout) | base_len:18 (ready to break) | RelVol:1.7x (strong buying) | ATR10:4.5% (healthy volatility) | ADX:26 (strong trend) | prox52:-3% (near highs)"
```

• BUY = No:

```
reason: "pivot_clear:-2.1% (below resistance) | base_len:7 (early
consolidation) | RelVol:0.8x (quiet volume) | ATR10:2.3% (low
volatility) | ADX:18 (weak trend) | prox52:-12% (far from highs)"
```

Decision Tree Summary

EOD Snapshot (scores/daily) → Compute Indicators

```
|
|— score < 70 ..... → WATCH
|
|— all gates pass ..... → BUY=Yes
|
|— any gate fails ..... → BUY=No (with reason)
```

Example Scenarios

Situation	Indicators	Result
Early breakout attempt	pivot_clear:+0.8% (testing breakout), base_len:8 (early consolidation), ADX:19 (weak trend), RelVol:1.0x (average)	BUY=No – needs more strength
Confirmed breakout	pivot_clear:+2.5% (clean breakout), base_len:18 (ready to break), RelVol:1.7x (strong buying), ADX:26 (strong trend), ATR:4.2% (healthy)	BUY=Yes – solid setup
Overheated run	pivot_clear:+8% (overextended), ATR:9% (too volatile), ADX:38 (euphoric)	BUY=No – wait for pullback

Configurable Parameters (default.yaml)

The BUY decision logic and several scoring thresholds are **configurable via** `config/default.yaml`. These parameters allow dynamic tuning of risk, breakout sensitivity, and volatility tolerance without code changes.

What's currently in your `default.yaml` (affects BUY / management)

```
rules:
  breakeven_gain_pct: 5.0
  # Move stop to breakeven after this gain %
  atr_chand_mult: 2.0
  # Chandelier/trailing stop: ATR multiple (normal regime)
  atr_chand_mult_euphoria: 1.4      # Tighter trailing stop in euphoria
  atr_init_mult: 2.0               # Initial stop distance on entry (in
```

ATRs)

```
euphoria:
  rsi_min: 75          # RSI threshold for euphoria regime
  adx_min: 30          # ADX threshold for euphoria regime
  alt_rsi_min: 70      # Alternative RSI gate if ADX is lower
  alt_adx_min: 25      # Alternative ADX gate if RSI is higher
  adx_slope5_min: 0    # ADX must be rising over last 5 bars
  soft_gates:          # Pre-filters before BUY logic is
applied
  min_score: 35
  min_relvol20: 0.8
  min_adx14: 22
  min_prox52w_pct: -10
  require_base_lenBars: 15
  breakout_overrides_soft_gates: true
```

What these do: - `breakeven_gain_pct`, `atr_*` → **position management**, not entry; they affect **sell/stop** alerts and **trim** behavior (and will influence the narrative added to `reason` for active trades). - `euphoria.*` → defines **overheated market**; when met, stop/trailing logic tightens and `reason` may include *"euphoric momentum"* phrasing. - `soft_gates.*` → **pre-filters**. If these aren't met, BUY won't even be evaluated (you'll see `reason` phrases like *"quiet volume"*, *"weak trend"*, *"early consolidation"* based on these thresholds).

Coverage check — what's missing for full BUY configurability

Your BUY gates currently used by the engine include additional thresholds that are **not** present in the YAML. To make BUY fully configurable (and to keep `reason` text aligned with config), add the following keys:

```
rules:
  buy:
    score_min: 70          # Overall quality floor for BUY
    prox52w_min_pct: -8    # Must be within -8% of 52W high
    pivot_clear_min_pct: 1.0 # Breakout must clear pivot by at least
1%                          #
    pivot_clear_max_pct: 5.0 # But not be overextended beyond 5%
    base_len_minBars: 15    # Mature base requirement
    relvol20_min: 1.5       # Volume confirmation
    adx14_min: 22           # Trend strength confirmation
    atr10_min_pct: 3.0      # Tradable volatility floor
    atr10_max_pct: 7.0      # Volatility ceiling (avoid whipsaw)
    day_change_max_pct: 6.0 # Avoid euphoric, chase-y candles
    liquidity_min_value: 50000000 # ₹5 Cr minimum traded value
    eod_only: true          # Enforce EOD snapshot for BUY decisions
```

Why add these? They are the exact numeric gates the BUY engine uses. Moving them to YAML ensures the BUY flag, alerts, and the `reason` text all come from a single, inspectable source of truth.

How config feeds the `reason` string

For each indicator, the renderer prints **stat** followed by **mapped language** based on the configured ranges:

- **Pivot clear** → uses `pivot_clear_min_pct` / `pivot_clear_max_pct`
 - `< 0%` → (below resistance)
 - `0-min%` → (testing breakout)
 - `min-max%` → (clean breakout)
 - `> max%` → (overextended)
- **Base length** → compares to `base_len_minBars`
 - `< base_len_min` → (early/immature base)
 - `≥ base_len_min` → (ready to break)
- **RelVol20** → compares to `relvol20_min`
 - `< min` → (quiet, low volume)
 - `≥ min` → (strong buying)
- **ATR10%** → uses `atr10_min_pct` / `atr10_max_pct`
 - `< min` → (low volatility)
 - `min-max` → (healthy tradable)
 - `> max` → (too volatile)
- **ADX14** → uses `adx14_min`
 - `< min` → (weak/sideways trend)
 - `≥ min` → (trend confirmed)
- **Proximity to 52W high** → uses `prox52w_min_pct`
 - `< min` → (far from highs / building base)
 - `≥ min` → (near breakout zone)
- **Score** → uses `score_min`
 - `< min` → (overall strength low)
 - `≥ min` → (quality met)
- **Day change %** → uses `day_change_max_pct`
 - `> max` → (too extended today)
- **Liquidity** → compares to `liquidity_min_value`
 - `< min` → (illiquid / avoid large slippage)

Example BUY=No (auto-composed):

```
reason: "pivot_clear: +0.3% (testing breakout) | base_len: 12 (early base) |  
RelVol: 1.2x (average volume) | ATR10: 2.7% (low volatility) | ADX: 20 (weak  
trend) | prox52: -9% (near but not yet) | Score: 68 (strength slightly low)"
```

Example BUY=Yes:

```
reason: "pivot_clear: +2.1% (clean breakout) | base_len: 18 (ready to break)  
| RelVol: 1.8x (strong buying) | ATR10: 4.1% (healthy tradable) | ADX: 26  
(trend confirmed) | prox52: -3% (near highs) | Score: 74 (quality met)"
```

The **same YAML thresholds** feed both the **BUY boolean** and the **human-language tags** — so the reason is always consistent with the decision.

Implementation Guide (quick)

1. **Load config** at app start and expose a typed `BuyConfig` object.
 2. Replace hard-coded thresholds in the BUY evaluator with `BuyConfig` values.
 3. Pass `BuyConfig` into the `reason` renderer to map stats → language using the same ranges.
 4. Include `config_version` in outputs so logs and emails indicate which YAML produced the decision.
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Summary

- **Scoring folder** structure clearly separates **daily (EOD)** vs **intraday** data.
- **Base_len_bars** counts daily consolidation bars from historical OHLC parquet.
- **Reason** field now shows both **stats and human-friendly meaning** side by side.
- This dual-format output makes reports self-explanatory for traders and analysts alike.